Notes: Gompers (JF, 1995)
Optimal Investment, Monitoring, and the Staging of Venture Capital

Contents

1 Big picture

- **Research question.** Why do venture capitalists invest in discrete stages rather than giving entrepreneurs all capital up front, and what determines the timing and size of those staged investments?
- **Central mechanism.** Staging gives the venture capitalist a repeated option to collect information, monitor the entrepreneur, and stop financing projects whose prospects deteriorate.
- **Agency problem.** Entrepreneurs have private information and private benefits, so they may continue negative-NPV projects or spend on strategies whose private return exceeds the investor's return.
- **Main proxy.** The duration of a financing round is a proxy for monitoring intensity: shorter duration means the VC comes back sooner to re-evaluate the firm.
- **Predictions.** Staging should be tighter when assets are less tangible, growth options are high, and R&D or asset specificity is high.
- **Main result.** Using 794 venture-backed firms and 2,143 financing rounds, the paper finds that financing duration declines when agency-cost proxies rise.

2 Incremental contribution

- The paper provides one of the earliest large-sample empirical tests of staged venture capital financing.
- It turns the time between financing rounds into a measurable governance variable.
- It links agency theory, real-options logic, and VC contract timing.
- It uses a random Venture Economics sample rather than only successful IPOs.
- It studies duration, round size, total funding, and number of rounds together.
- It documents that VC market liquidity changes staging behavior, opening the capital-supply/free-cash-flow question in venture finance.

3 Data and measurement

The core data are a random sample of 794 venture capital-backed firms selected from Venture Economics, with 2,143 financing rounds from 1961 to July 1992. Firms are included if first financing occurred before January 1, 1990. Outcomes are IPO, merger/acquisition, liquidation/bankruptcy, and still private. Because private-firm accounting data are unavailable, the paper matches firms to COMPUSTAT industry averages to proxy for asset tangibility, market-to-book, and R&D intensity.

Detailed interpretation. Duration between financing rounds is the key empirical measure of monitoring intensity. Industry proxies are noisy but make the theory testable across private firms.

4 Models and empirical specifications

4.1 Staging as monitoring

$$\text{Shorter duration between rounds} \Rightarrow \text{more frequent monitoring.}$$

Expected agency costs rise when tangibility falls, growth options rise, and R&D intensity rises.

Detailed interpretation. Low tangibility, high market-to-book, and high R&D all make inefficient continuation more costly, so the VC should monitor more often.

4.2 Hazard-rate formulation

$$h(t) = \frac{\Pr(\text{receiving funding between } t \text{ and } t + \Delta t)}{\Pr(\text{receiving funding after } t)}.$$

Detailed interpretation. A duration model is appropriate because some firms do not receive another observed round due to IPO, acquisition, liquidation, or censoring.

4.3 Empirical regressions

$$Duration_{ir} = f(Stage_{ir}, VCCommitments_{t-1}, Tangibility_{jt}, MTB_{jt}, R\&D_{jt}, Age_{ir}, Amount_{ir}) + \varepsilon_{ir}.$$

$$\log(Amount_{ir}) = \alpha + \beta X_{ir} + \varepsilon_{ir}.$$

$$\log(TotalFunding_i) = \alpha + \gamma Outcome_i + \delta Industry_i + \eta NumberRounds_i + \varepsilon_i.$$

Detailed interpretation. The duration regressions test monitoring intensity. Round-size and total-funding regressions test how capital provision responds to information and scale.

5 Identification and empirical design

The paper is not a quasi-experimental design. It is a theory-guided empirical test. Identification comes from cross-sectional variation in firm stage, industry tangibility, market-to-book, R&D intensity, firm age, outcomes, and time-series variation in VC commitments. The main threats are unobserved project quality, missing non-VC financing, noisy public-industry proxies for private firms, and unobserved VC heterogeneity.

6 Section-by-section study guide

- **Introduction:** staging as monitoring and option-to-abandon.
- **Theory section:** agency-cost predictions for duration and amount.
- **Data section:** Venture Economics sample, outcomes, stage classifications, industry proxies.
- **Empirical section:** duration, amount, total funding, and number-of-rounds regressions.
- **Conclusion:** staging is a governance technology, not merely financing convenience.

7 Tables and figures

7.1 Figure 1: New commitments to venture capital funds in constant 1993 dollars + Table I: Time Series of Random Sample from the Venture Economics Database

Read: Figure 1 shows the venture capital commitment boom of the early-to-mid 1980s. Table I shows that the sample becomes much denser after the early 1980s, with rounds rising to 234 by 1989.

Detailed interpretation. These visuals establish the capital-supply environment behind the liquidity tests.

Economic message. VC fund-raising cycles can change monitoring frequency and investment size.

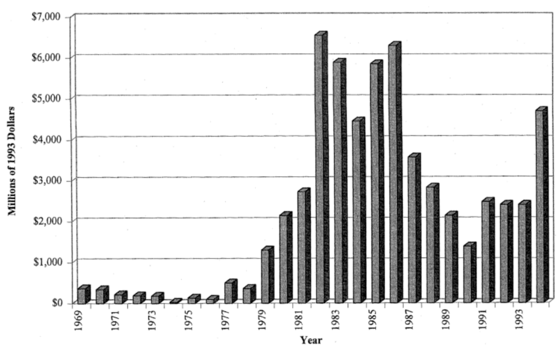

Figure 1. New commitments to venture capital funds in constant 1993 dollars.

(a) New VC commitments

Table I

Time Series of Random Sample from the Venture Economics Database

The sample is 794 randomly selected companies from the set of firms that received their first venture capital investment prior to January 1, 1990. The table shows the number of rounds, total amount invested, and the number of new firms in each year in the sample of random firms. Amount of known investment is in thousands of dollars.

Year	Rounds of Venture Capital Financing	Amount of Venture Capital Investment (Thousands of Dollars)	Number of New Firms Receiving Venture Capital
1961	1	280	1
1962	1	200	1
1968	1	250	1
1969	3	2,135	2
1970	6	1,911	4
1971	8	2,257	4
1972	4	2,470	1
1973	9	19,436	6
1974	9	16,384	1
1975	14	10,775	9
1976	17	18,174	9
1977	23	9,064	12
1978	35	17,733	24
1979	44	64,788	30
1980	55	34,392	30
1981	77	113,794	50
1982	126	179,965	64
1983	170	380,648	82
1984	208	518,408	84
1985	179	517,447	66
1986	204	484,405	95
1987	219	434,966	60
1988	225	594,322	85
1989	234	418,940	67
1990	142	157,786	0
1991	109	142,878	0
1992	16	42,771	0

(b) Time-series sample table

7.2 Table II: Percentage of Investment by Industry and Stage of Development in Each Year

Read: VC investments are concentrated in technology-intensive sectors and shift gradually from early-stage to later-stage investments.

Detailed interpretation. This supports the premise that VC is concentrated where information and monitoring problems are severe.

Economic message. Venture capital targets industries where conventional financing faces high information frictions.

7.3 Table III: Outcomes for 794 Venture Capital-Backed Firms by Industry

Read: The outcome distribution is 127 IPOs, 134 mergers/acquisitions, 88 liquidations/bankruptcies, and 215 still-private firms.

Detailed interpretation. Outcome variation allows the paper to test whether continued financing is associated with successful projects.

Economic message. Staging reallocates capital toward firms that pass repeated information screens.

Table II

Percentage of Investment by Industry and Stage of Development in Each Year

Data are 2143 financing rounds for a random sample of 794 venture capital-backed firms. Panel A shows the industry composition of venture investments in the sample through time. Industry classifications are reported by Venture Economics. Panel B shows how the stage of firm development for venture investments varies in the sample. Early stage investments are seed, startup, early, first, and other early stage investments. Late stage financing is second, third, or bridge stage investments. All values are in percent.

	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
Panel A: Percentage of Rounds Invested by Industry															
Communications	28.6	23.5	18.2	8.6	18.2	11.1	13.3	12.9	14.9	13.3	12.4	12.3	11.9	12.0	11.5
Computers	0.0	0.0	0.0	2.9	2.3	3.7	4.0	8.9	10.1	10.1	6.2	3.9	4.6	4.4	1.3
Computer related	0.0	5.9	4.5	11.4	15.9	11.1	13.3	21.8	19.6	16.1	17.5	14.2	14.2	11.1	15.8
Computer software	7.1	0.0	0.0	0.0	0.0	0.0	6.7	10.5	13.7	9.6	14.7	13.2	11.4	9.3	11.1
Electronic components	0.0	0.0	4.5	0.0	2.3	5.6	2.7	0.8	3.6	3.2	4.5	3.4	4.6	3.6	3.0
Other electronics	28.6	35.3	4.5	2.9	.4.5	11.1	10.7	8.9	3.6	7.8	6.8	4.4	4.1	4.9	5.6
Biotechnology	7.1	0.0	0.0	2.9	6.8	3.7	2.7	5.6	3.6	3.7	1.1	4.4	5.0	6.7	6.4
Medical/health	0.0	0.0	9.1	20.0	4.5	9.3	2.7	6.5	10.1	11.0	16.4	13.2	14.6	15.6	12.0
Energy	14.3	0.0	31.8	5.7	6.8	7.4	4.0	3.2	3.0	0.0	0.6	1.5	0.5	1.3	0.9
Consumer products	0.0	11.8	4.5	17.1	13.6	18.5	10.7	4.8	6.0	11.9	9.6	9.3	11.4	15.1	13.7
Industrial products	7.1	17.6	4.5	14.3	9.1	9.3	17.3	9.7	4.2	6.0	4.0	7.8	7.8	9.8	8.5
Transportation	0.0	0.0	18.2	0.0	4.5	0.0	1.3	2.4	0.6	0.9	0.6	2.0	0.9	1.8	0.0
Other	7.1	5.9	0.0	14.3	11.4	9.3	10.7	4.0	7.1	6.4	5.6	10.3	9.1	4.4	10.3
Panel B: Percentage of Rounds Invested by Stage of Development															
Early stage	69.2	92.9	85.7	63.3	70.6	66.7	52.3	59.5	54.9	55.3	47.7	43.1	39.7	40.0	34.5
Late stage	30.8	7.1	14.3	36.7	29.4	33.3	47.7	40.5	45.1	44.7	52.3	56.9	60.3	60.0	65.5

Table III

Outcomes for 794 Venture Capital-Backed Firms by Industry

The number of firms that had performed an initial public offering, merged, went bankrupt, or remained private as of July 31, 1992. The first column is firms that went public. The second column is all firms that were acquired or merged with another company. The third column is all firms that filed for bankruptcy. The fourth column is all firms that are still private and have not received venture capital financing since January 1, 1988. Percentage of outcome classification for each industry are in parentheses.

Industry	IPOs	Mergers/ Acquisitions	Liquidations/ Bankruptcies	Private
Communications	17	17	9	26
	(24.6)	(24.6)	(13.0)	(37.7)
Computers	5	7	9	4
	(20.0)	(28.0)	(36.0)	(16.0)
Computer related	20	19	15	15
	(29.0)	(27.5)	(21.7)	(21.7)
Computer software	11	9	11	20
	(21.6)	(17.6)	(21.6)	(39.2)
Electronic components	2	6	0	9
	(11.8)	(35.3)	(0.0)	(52.9)
Other electronics	6	9	6	9
	(20.0)	(30.0)	(20.0)	(30.0)
Biotechnology	9	5	2	2
	(50.0)	(27.8)	(11.1)	(11.1)
Medical/health	17	17	12	10
	(30.4)	(30.4)	(21.4)	(17.9)
Energy	4	3	2	11
	(20.0)	(15.0)	(10.0)	(55.0)
Consumer products	19	10	6	33
	(27.9)	(14.7)	(8.8)	(48.5)
Industrial products	4	20	6	27
	(7.0)	(35.1)	(10.5)	(47.4)
Transportation	5	2	1	4
	(41.7)	(16.7)	(8.3)	(33.3)
Other	8	10	9	45
	(11.1)	(13.9)	(12.5)	(62.5)
Total	127	134	88	215
	(22.5)	(23.8)	(15.6)	(38.1)

7.4 Table IV: Number of Investments, Age at First Funding, and Total Funding Received by Industry and Outcome

Read: IPO firms generally receive more rounds and more total capital than firms that are liquidated or acquired.

Detailed interpretation. The pattern is consistent with VCs continuing projects after favorable information arrives.

Economic message. VC investment is a sequence of continuation decisions.

7.5 Table V: Duration, Amount of Investment, and Cash Utilization by Stage of Development

Read: Early rounds are smaller; later rounds tend to be larger and have higher cash utilization.

Detailed interpretation. Early rounds buy information; later rounds finance expansion after uncertainty is partially resolved.

Economic message. Staging combines learning with scale.

Table IV

Number of Investments, Age at First Funding, and Total Funding Received by Industry and Outcome

The sample is 794 venture capital-backed firms randomly selected from the Venture Economics database. The number of rounds, age at first funding, and total venture capital financing (in constant 1992 dollars) are tabulated for various industries and various outcomes. Firms can either go public in an IPO, go bankrupt, or be acquired. Average total funding is in thousands of dollars. Average age at first funding is in years. Median values are in parentheses.

	Number of Rounds				Age at First Funding			
Industry	Full	IPO	Bankrupt	Acquired	Full	IPO	Bankrupt	Acquired
Communications	2.78	3.41	2.44	2.47	3.46	3.29	2.56	5.11
	(2)	(2)	(2)	(2)	(0.92)	(1.34)	(1.87)	(1.34)
Computers	3.89	4.60	4.33	3.42	4.19	1.11	2.75	2.30
	(3)	(6)	(4)	(3)	(1.33)	(0.17)	(1.84)	(1.29)
Computer related	3.66	4.0	3.47	3.32	4.29	3.74	4.10	4.10
	(3)	(4)	(2)	(3)	(1.88)	(1.67)	(2.75)	(2.75)
Computer software	2.99	2.91	2.00	3.22	3.59	3.83	4.30	4.30
	(3)	(2)	(1)	(3)	(1.92)	(3.67)	(2.59)	(2.59)
Electronic components	3.27	4.00	na	3.50	0.86	0.53	na	0.77
	(2)	(4)		(3.5)	(0.00)	(0.53)		(0)
Other electronics	3.21	2.50	3.50	2.78	3.45	2.54	5.67	5.46
	(2)	(2)	(2.5)	(2)	(2.38)	(3.17)	(3)	(3.92)
Biotechnology	3.69	3.56	4.00	4.60	1.21	0.89	0.37	2.08
	(4)	(3)	(4)	(4)	(0.71)	(0.50)	(0.37)	(2)
Medical/health	2.98	3.94	1.91	2.53	1.97	2.30	0.58	1.59
	(2)	(3)	(1)	(2)	(1.00)	(1.41)	(0.12)	(0.71)
Energy	1.91	2.25	2.00	1.67	2.85	7.01	2.13	2.00
	(1)	(2)	(2)	(1)	(2.00)	(5.17)	(2.13)	(2)
Consumer products	2.14	2.16	2.33	1.20	5.90	7.63	0.98	17.92
	(1)	(2)	(2)	(1)	(1.67)	(4.41)	(0.75)	(17.35)
Industrial products	2.09	3.75	2.17	1.65	3.79	5.94	18.97	4.66
	(1)	(2)	(1.5)	(1)	(2.25)	(6.38)	(10.46)	(2.46)
Transportation	1.93	2.00	2.00	2.50	6.33	15.84	na	9.09
	(2)	(2)	(2)	(2.5)	(5.67)	(5.27)		(9.09)
Other	1.60	1.63	1.78	1.80	5.83	12.48	1.00	10.11
	(1)	(1)	(1)	(1)	(2.25)	(3.17)	(0.46)	(5.96)

Table IV—*Continued*

	Total Funding (Thousands of Dollars)				Number of Firms			
Industry	Full	IPO	Bankrupt	Acquired	Full	IPO	Bankrupt	Acquired
Communications	7,402	7,017	2,841	4,693	98	17	9	17
	(3,300)	(4,260)	(2,000)	(1,447)				
Computers	16,162	20,483	15,507	5,363	27	5	9	7
	(7,750)	(20,483)	(6,000)	(2,492)				
Computer related	8,062	13,386	7,432	4,965	90	20	15	19
	(4,050)	(8,766)	(5,000)	(4,323)				
Computer software	4,537	7,584	2,978	6,108	77	11	11	9
	(2,092)	(3,463)	(1,500)	(1,604)				
Electronic components	10,479	12,425	na	6,914	22	2	0	6
	(3,484)	(12,425)		(2,483)				
Other electronics	5,228	6,371	6,777	3,160	41	6	6	9
	(4,000)	(6,330)	(1,637)	(1,875)				
Biotechnology	8,562	12,716	7,659	8,066	29	9	2	5
	(5,500)	(10,957)	(5,716)	(12,000)				
Medical/health	5,680	10,246	2,853	3,736	90	17	12	17
	(3,000)	(3,645)	(1,236)	(3,400)				
Energy	3,086	3,918	4,698	1,963	22	4	2	3
	(899)	(2,434)	(4,698)	(850)				
Consumer products	6,551	11,161	2,654	5,694	103	19	6	10
	(2,237)	(5,473)	(1,259)	(958)				
Industrial products	2,982	9,855	3,149	2,274	89	4	6	20
	(1,500)	(7,875)	(1,881)	(1,200)				
Transportation	4,983	6,468	4,000	5,875	15	5	1	2
	(3,252)	(2,500)	(4,000)	(5,875)				
Other	4,526	12,889	2,810	8,738	96	8	9	10
	(1,968)	(8,000)	(1,664)	(4,277)				

Table V

Duration, Amount of Investment, and Cash Utilization by Stage of Development

The sample is 794 venture capital-backed firms randomly selected from the Venture Economics database. Investment type is self-reported stage of development for venture capital-backed firms at time of investment. Median values are in parentheses. Time to next funding is the duration (in years) from one reported financing round to the next. Amount of funding is the average size of a given type of financing round (in thousands of 1992 dollars). Cash utilization is the rate at which the firm is using cash between rounds of financing (in thousands of 1992 dollars per year).

Type of Funding	Time to Next Funding	Amount of Funding (Thousands of Dollars)	Cash Utilization (Thousands of Dollars per Year)	Number
Seed	1.63	921	565	122
	(1.17)	(290)	(248)	
Startup	1.21	2,387	1,987	129
	(1)	(1,098)	(1,098)	
Early stage	1.03	1,054	1,023	114
	(0.83)	(750)	(904)	
First stage	1.08	1,928	1,785	288
	(0.92)	(1,000)	(1,087)	
Other early	1.08	2,182	2,020	221
	(0.75)	(1,200)	(1,600)	
Expansion	1.26	2,343	1,860	377
	(0.88)	(1,000)	(1,136)	
Second stage	1.01	2,507	2,482	351
	(0.83)	(1,350)	(1,627)	
Third stage	0.86	2,784	3,237	181
	(0.75)	(1,200)	(1,600)	
Bridge	0.97	2,702	2,785	454
	(0.83)	(1,500)	(1,807)	

Table V

7.6 Table VI: Regressions for Duration and Funding Amount per Round Controlling for Firm and Industry Factors

Read: Tangibility lengthens duration, market-to-book and R&D shorten duration, and prior-year VC commitments shorten duration and increase round size.

Detailed interpretation. This is the central evidence that agency-cost proxies determine monitoring intensity.

Economic message. Staging is organized around expected agency costs and capital supply.

Table VI

Regressions for Duration and Funding Amount per Round Controlling for Firm and Industry Factors

The sample is 2143 funding rounds for 794 venture capital-backed firms for the period 1961 to 1992. The dependent variables are the time in years from funding date to the next funding date and the logarithm of the round's funding amount in thousands of 1992 dollars. Independent variables include a dummy variable that equals 1 if the funding round is either seed or startup (early stage) and a dummy variable that equals 1 if the round is either early, first, or other early (middle stage). Liquidity in the venture capital industry is controlled using new capital commitments to venture capital partnerships in the previous year in constant 1992 dollars. Tangibility of assets is measured by the average ratio of tangible assets to total assets for company's in the firm's industry. Market-to-book is the average industry ratio of market value of equity to book value of equity. Research and development intensity is proxied by the average industry ratios of R&D to sales or R&D to assets. The age of the venture capital-backed firm is months from incorporation to financing date. Panel A are maximum-likelihood estimates for Weibull distribution duration models. Panel B estimates are ordinary least squares. *t*-statistics for coefficients are in parentheses.

Panel A: Regressions for Duration of Financing Round

	Dependent Variable: Duration of Financing Round					
Independent Variables	(1)	(2)	(3)	(4)	(5)	(6)
Constant	−0.030	0.361	0.407	0.417	0.070	0.082
	(−0.19)	(2.50)	(2.86)	(2.93)	(0.39)	(0.42)
Investment was in an early stage firm?	0.051	0.040	0.037	0.047	0.036	0.031
	(0.63)	(0.49)	(0.44)	(0.56)	(0.44)	(0.38)
Investment was in a middle stage firm?	−0.054	−0.058	−0.103	−0.094	−0.102	−0.106
	(−0.93)	(−1.00)	(−1.72)	(−1.55)	(−1.71)	(−1.75)
Capital committed to new venture funds in previous year	−0.60 × E-04	−0.56 × E-04	−0.52 × E-04	−0.55 × E-04	−0.54 × E-04	−0.56 × E-04
	(−4.97)	(−4.56)	(−4.10)	(−4.41)	(−4.22)	(−4.36)
Industry ratio of tangible assets to total assets	0.405				0.400	0.398
	(4.01)				(3.84)	(3.23)
Industry market-to-book ratio		−0.047			0.000	−0.019
		(−1.87)			(0.00)	(−0.41)
Industry ratio of R&D expense to sales			−3.390		−2.268	
			(−2.52)		(−1.79)	
Industry ratio of R&D expense to total assets				−0.795		−0.194
				(−2.69)		(−1.67)
Age of the firm at time of venture financing round	0.016	0.016	0.016	0.017	0.016	0.017
	(3.58)	(3.68)	(3.49)	(3.56)	(3.52)	(3.60)
Logarithm of the amount of venture financing this round	0.011	0.012	0.016	0.011	0.010	0.009
	(0.71)	(0.73)	(0.68)	(0.65)	(0.59)	(0.52)
Pseudo-R^2	0.045	0.037	0.045	0.046	0.053	0.051
Model χ^2	56.00	42.25	48.64	49.47	62.74	60.38

Table VI—*Continued*

Panel B: Regressions for Size of Each Financing Round

	Dependent Variable: Logarithm of the Financing Amount in the Round					
Independent Variables	(1)	(2)	(3)	(4)	(5)	(6)
Constant	6.580	6.756	6.902	6.929	6.379	6.108
	(38.00)	(39.02)	(46.39)	(46.27)	(26.97)	(22.39)
Investment was in an early stage firm?	−0.635	−0.608	−0.703	−0.703	−0.748	−0.760
	(−4.14)	(−4.22)	(−4.83)	(−4.83)	(−5.14)	(−5.23)
Investment was in a middle stage firm?	−0.224	−0.216	−0.308	−0.309	−0.314	−0.328
	(−2.29)	(−2.20)	(−3.06)	(−3.06)	(−3.13)	(−3.26)
Capital committed to new venture funds in previous year	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
	(3.94)	(4.21)	(3.91)	(3.98)	(3.10)	(3.08)
Industry ratio of tangible assets to total assets	0.352				0.612	0.810
	(2.23)				(3.64)	(4.16)
Industry market-to-book ratio		−0.051			0.041	0.084
		(−0.64)			(0.49)	(1.03)
Industry ratio of R&D expense to sales			1.578		3.618	
			(0.72)		(1.56)	
Industry ratio of R&D expense to total assets				−0.099		1.372
				(−0.21)		(2.43)
Age of the firm at time of venture financing round	−0.019	−0.019	−0.014	−0.014	−0.014	−0.014
	(−2.58)	(−2.58)	(−1.82)	(−1.86)	(−1.90)	(−1.89)
R^2	0.031	0.028	0.039	0.031	0.041	0.044
F-statistic	9.33	8.40	8.50	8.40	8.03	8.55

7.7 Table VII: Regression for Total Venture Capital Funding and Number of Rounds of Financing

Read: IPO firms receive more total funding and more financing rounds.

Detailed interpretation. Successful projects survive more screens and accumulate more capital.

Economic message. The VC process shifts resources toward projects revealed to be promising.

7.8 Table VIII: Regressions for Duration with the Sample Split into High Technology and Low Technology Companies

Read: The staging results appear in both high-technology and low-technology subsamples.

Detailed interpretation. The agency-cost logic is not purely a high-tech artifact.

Economic message. Staged monitoring is a general VC governance response.

Table VII

Regression for Total Venture Capital Funding and Number of Rounds of Financing

The sample is 794 venture capital-backed firms for the period 1961 to 1992. The dependent variables are the total venture capital funding that the firm received in thousands of 1992 dollars and the number of distinct rounds of venture financing. Independent variables include a dummy variable that equals 1 if the firm completed an initial public offering, a dummy variable that equals 1 if the firm filed for bankruptcy, and a dummy variable that equals 1 if the firm was acquired by or merged with another company. Tangibility of assets is measured by the average ratio of tangible assets to total assets for companies in the firm's industry. Market-to-book is the average industry ratio of market value of equity to book value of equity. Research and development intensity is proxied by the average industry ratios of R&D to sales or R&D to assets. Estimates in Panel A are from ordinary least squares regressions. Estimates for equations in Panel B are from Poisson regressions. *t*-statistics for regression coefficients are in parentheses.

Panel A: Regressions for Total Funding

	Dependent Variable: Logarithm of Total Venture Financing Received					
Independent Variables	(1)	(2)	(3)	(4)	(5)	(6)
Constant	11.017	7.075	7.290	7.427	4.714	4.771
	(12.38)	(38.22)	(67.40)	(68.38)	(4.41)	(4.47)
Firm exited via an IPO?	1.043	1.018	0.882	0.905	0.664	0.666
	(6.18)	(6.01)	(4.88)	(4.96)	(4.31)	(4.32)
Firm went bankrupt or was liquidated?	−0.023	−0.024	−0.102	−0.047	−0.085	−0.077
	(−0.11)	(−0.12)	(−0.49)	(−0.22)	(−0.48)	(−0.44)
Firm exited via merger or acquisition?	−0.125	−0.129	−0.003	−0.009	0.124	0.123
	(−0.76)	(−0.78)	(−0.02)	(−0.05)	(0.84)	(0.83)
Industry ratio of tangible assets to total assets	−3.660				1.118	1.036
	(−3.90)				(1.12)	(1.04)
Industry market-to-book ratio		0.311			0.402	0.420
		(2.90)			(3.52)	(3.69)
Industry ratio of R&D expense to sales			13.033		3.600	
			(4.02)		(1.24)	
Industry ratio of R&D expense to total assets				5.709		2.540
				(1.95)		(1.02)
Number of rounds of venture financing received					0.396	0.399
					(15.19)	(15.43)
R^2	0.073	0.064	0.067	0.048	0.337	0.336
F-statistic	13.40	11.60	11.05	7.82	44.37	44.26

Table VII—*Continued*

Panel B: Poisson regressions for number of rounds

	Dependent Variable: Number of Financing Rounds Received			
Independent Variables	(1)	(2)	(3)	(4)
Constant	2.904	0.945	0.796	0.888
	(10.51)	(13.88)	(18.82)	(21.55)
Firm exited via an IPO?	0.255	0.239	0.186	0.203
	(4.35)	(4.06)	(2.91)	(3.18)
Firm went bankrupt or was liquidated?	0.054	0.052	0.017	0.052
	(0.71)	(0.68)	(0.23)	(0.68)
Firm exited via merger or acquisition?	0.027	−0.006	−0.010	−0.004
	(0.43)	(−0.09)	(−0.16)	(−0.06)
Industry ratio of tangible assets to total assets	−2.054			
	(−6.96)			
Industry market-to-book ratio		0.021		
		(0.54)		
Industry ratio of R&D expense to sales			7.416	
			(6.25)	
Industry ratio of R&D expense to total assets				2.907
				(2.73)
Pseudo-R^2	0.020	0.006	0.019	0.007
Model χ^2	60.25	18.55	50.07	18.81

expected agency costs rise.

Table VIII

Regressions for Duration Controlling for Firm and Industry Factors with the Sample Split into High Technology and Low Technology Companies

The sample is 2143 funding rounds for 794 venture capital-backed firms for the period 1961 to 1992. The dependent variable is the time in years from funding date to the next funding date. Independent variables include a dummy variable that equals 1 if the funding round is either seed or startup (early stage) and a dummy variable that equals 1 if the round is either early, first, or other early (middle stage). Liquidity in the venture capital industry is controlled using new capital commitments to venture capital partnerships in the previous year in constant 1992 dollars. Tangibility of assets is measured by the average ratio of tangible assets to total assets for company's in the firm's industry. Market-to-book is the average industry ratio of market value of equity to book value of equity. Research and development intensity is proxied by the average industry ratios of R&D to sales or R&D to assets. The age of the venture capital-backed firm is months from incorporation to financing date. All regressions are maximum-likelihood estimates for Weibull distribution duration models. *t*-statistics for coefficients are in parentheses.

Panel A: Regressions for Duration of Financing Round for High Technology Industries

	Dependent Variable: Duration of Financing Round					
Independent Variables	(1)	(2)	(3)	(4)	(5)	(6)
Constant	−0.036	0.464	0.542	0.549	0.022	−0.056
	(−0.17)	(2.56)	(2.99)	(3.03)	(0.09)	(−0.22)
Investment was in an early stage firm?	0.066	0.065	0.042	0.046	0.036	0.029
	(0.68)	(0.66)	(0.42)	(0.45)	(0.36)	(0.30)
Investment was in a middle stage firm?	−0.022	−0.009	−0.085	−0.084	−0.101	−0.109
	(−0.32)	(−0.13)	(−1.15)	(−1.14)	(−1.38)	(−1.48)
Capital committed to new venture funds in previous year	−0.63 × E-04	−0.59 × E-04	−0.54 × E-04	−0.57 × E-04	−0.59 × E-04	−0.61 × E-04
	(−4.50)	(−4.01)	(−3.59)	(−3.82)	(−3.81)	(−3.93)
Industry ratio of tangible assets to total assets	0.514				0.553	0.633
	(3.47)				(3.60)	(3.55)
Industry market-to-book ratio		−0.059			0.037	−0.003
		(−0.66)			(0.39)	(−0.03)
Industry ratio of R&D expense to sales			−2.418		−0.986	
			(−1.83)		(−0.51)	
Industry ratio of R&D expense to total assets				−0.541		−0.315
				(−1.88)		(−1.68)
Age of the firm at time of venture financing round	−0.001	−0.001	−0.005	−0.005	−0.006	−0.005
	(−0.17)	(−0.09)	(−0.70)	(−0.70)	(−0.74)	(−0.64)
Logarithm of the amount of venture financing this round	0.001	0.003	−0.002	−0.002	−0.004	−0.005
	(0.07)	(0.13)	(−0.08)	(−0.08)	(−0.21)	(−0.22)
Pseudo-R^2	0.036	0.029	0.032	0.032	0.041	0.040
Model χ^2	31.07	20.75	21.85	21.99	33.63	33.81

Table VIII—*Continued*

Panel B: Regressions for Duration of Financing Round for Low Technology Industries

	Dependent Variable: Duration of Financing Round					
Independent Variables	(1)	(2)	(3)	(4)	(5)	(6)
Constant	−0.078	0.399	0.382	0.400	0.130	0.208
	(−0.29)	(1.50)	(1.54)	(1.62)	(0.45)	(0.62)
Investment was in an early stage firm?	−0.050	−0.095	−0.087	−0.068	−0.093	−0.098
	(−0.35)	(−0.66)	(−0.61)	(−0.48)	(−0.65)	(−0.69)
Investment was in a middle stage firm?	−0.148	−0.194	−0.210	−0.179	−0.196	−0.195
	(−1.45)	(−1.88)	(−2.03)	(−1.72)	(−1.89)	(−1.88)
Capital committed to new venture funds in previous year	−0.43 × E-04	−0.38 × E-04	−0.37 × E-04	−0.40 × E-04	−0.38 × E-04	−0.38 × E-04
	(−1.86)	(−1.89)	(−1.68)	(−1.72)	(−1.68)	(−1.62)
Industry ratio of tangible assets to total assets	0.433				0.393	0.337
	(2.90)				(2.55)	(2.17)
Industry market-to-book ratio		−0.076			−0.053	−0.078
		(−1.41)			(−0.91)	(−1.43)
Industry ratio of R&D expense to sales			−4.270		−2.006	
			(−2.18)		(−1.89)	
Industry ratio of R&D expense to total assets				−1.067		−0.388
				(−2.41)		(−1.66)
Age of the firm at time of venture financing round	0.026	0.028	0.030	0.031	0.029	0.030
	(4.07)	(4.34)	(4.44)	(4.54)	(4.34)	(4.41)
Logarithm of the amount of venture financing this round	0.022	0.018	0.017	0.014	0.019	0.017
	(0.79)	(0.63)	(0.61)	(0.50)	(0.66)	(0.59)

8 Result map

- VC investment concentrates in early-stage and high-tech settings.
- Duration is shorter when asset tangibility is lower, growth options are higher, and R&D intensity is higher.
- Round sizes are smaller for early-stage companies and larger when assets are more tangible.
- IPO firms receive more funding and more rounds.
- VC market liquidity increases round size and reduces duration.

9 Short summary

Gompers studies staged venture capital financing as a governance mechanism. The paper shows that VC staging responds to agency-cost proxies and that successful firms receive more financing rounds and total funding. The evidence supports the interpretation that VCs use staged capital infusions to monitor entrepreneurs, gather information, and preserve abandonment options.

10 Conclusion

10.1 Core conclusion

Staged venture capital financing is a central governance technology. It lets VCs finance risky startups while periodically reassessing whether the project should continue.

10.2 Economic intuition

Entrepreneurs have private information and private benefits. Giving all capital up front would weaken investor control. Staging keeps the entrepreneur on a tight leash by forcing repeated information revelation and renegotiation.

10.3 Incremental contribution

The paper turns the timing of venture financing into an empirical object and shows that the timing and size of rounds respond systematically to agency-cost and information proxies.

10.4 Limitations

The paper does not observe project quality at each round, non-VC financing, full contract terms, or VC fixed effects. The evidence is best interpreted as theory-consistent rather than causal.

10.5 Takeaway

Venture capital is not just money for startups. It is a repeated governance process that combines finance, monitoring, and real-options control.